
CULINARY DETECTIVE: IDENTIFYING A DISH’S REGION FROM ITS DESCRIPTION

A PREPRINT

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1 Observing Baseline Performance

Observing the baseline performance, we notice the following trends:

1. **GlobalTrekker** Average Score: 0.7517
2. **CulinaryDetective** Average Score: 0.3491
3. **LinguaLocale** Average Score: 0.8437

LinguaLocale appears to be the best performing section and CulinaryDetective the worst, with substantial room for improvement (the corresponding prompts can be found in the appendix).

1.1 Global Trekker Example

```
{
  "score": 0.0, "expected_country": "China", "expected_place": "Harbin",
  "pred_country": "Russia", "pred_place": "None",
  "question": {
    "type": "GlobalTrekker",
    "paragraph": "Every winter, a northern city along a wide river that freezes solid transforms into a temporary \u201cice city\u201d built from enormous blocks of river ice and packed snow. Full-size castles, temples, and towers are carved and lit from within by colorful lights, so visitors can wander glowing streets, slide down ice slides, and watch fireworks burst above neon-blue spires in temperatures far below freezing. What began as a small local ice-lantern fair has grown into one of the world\u2019s largest winter festivals, drawing sculptors and tourists from across the region to this famously frigid destination.",
    "country": "China", "city": "Harbin",
    "info": {"reference": "https://en.wikipedia.org/wiki/Harbin_International_Ice_and_Snow_Sculpture_Festival",
    "hints": "temporary \u201cice city\u201d, world\u2019s largest winter festivals, ice castles and temples"}
  }
}
```

This is a representative example of data ambiguity in the Global Trekker question. Here, the expected prediction for the given ice festival paragraph is the country of China and city of Harbin. However, our model chose Russia, with no particular city. The text of the paragraph never names China or Harbin, only ice city, frozen river, winter festival, all cues that fit Siberia’s climate as well. The miss in this case appears driven by an ambiguous description, and guided by the multiple references to ice, overshadowing the use of the term "ice-lantern".

1.2 Culinary Detective Example

```
{
  "idx": 63, "type": "CulinaryDetective", "score": 0.6,
  "expected_country": "Brazil", "expected_place": "All",
  "pred_country": "United States", "pred_place": "South".,
  "question": {
    "type": "CulinaryDetective",
    "ingredient": "black beans, ham, bacon, onion, garlic, bay leaves",
    "description": "Slow cooked black beans with assorted pork cutlets, hearty, salty",
    "country": "Brazil",
    "region": "All",
    "info": {
      "reference": "https://www.daringgourmet.com/feijoadada/"
    }
  }
}
```

In this example, we identify an incident in which RAG is being misleading. In this case, we expected the response: [Brazil], [All], identifying the country of Brazil and that the food is found in all regions of it. However, our model chose [United States], [South], indicating that the model predicts the food is eaten in the US. This is likely because the ingredients overlap heavily with U.S. Southern stew recipes, so the model latched onto the wrong cuisine/region. When we zero-shot this particular example, we get a prediction closer to the gold answer: [Brazil], [South]. Although the model is incorrect about the region, it did get the correct country, showing that removing the misleading retrieved context improved the prediction. Thus there is room for improvement on the RAG pipeline.

1.3 Lingua Locale

```
{
  "idx": 142, "type": "LinguaLocale", "score": 0.4,
  "expected_country": "Kenya", "expected_place": "",
  "pred_country": "Namibia", "pred_place": "none",
  "question": {
    "type": "LinguaLocale",
    "sentence": "Why school heads, Knut have ganged up against Junior School teachers",
    "country": "Kenya",
    "info": {
      "reference": "https://nation.africa/kenya/news/education/school-heads-knut-ganged-up-against-junior-school-teachers-5267358",
      "hints": "KNUT- Kenya National Union of Teachers"
    }
  }
}
```

In this example, the expected response is Kenya, but the model predicted Namibia. With English across many African countries, it is likely that the model defaulted to a different English-speaking African country instead of the gold, suggesting a language-cluster bias toward a frequent/default country when country-specific cues (e.g., “Kenya,” “Nairobi”) are absent.

2 Proposed Method to Improve

Inspecting our baseline results, we see that the average accuracy score of the CulinaryDetective is 34.91%, the lowest score of the three categories (and offering the greatest room for improvement). Thus, we will focus on the CulinaryDetective group of predictions, and we find that the missed predictions show both country drift and region drift.

We hypothesize that errors stem primarily from noisy or off-distribution retrieval (the Kaggle recipes are not equal to the evaluation set we use), and unconstrained region outputs. As a result, our RAG pipeline feeds irrelevant context and degrades predictions. As found in literature, RAG quality is sensitive to corpus noise (Lewis 2020; Cuconasu 2024) and better grounding yields gains (reddit.com/r/LangChain). We also note that prompt tweaks alone aren't moving the needle substantially and reliably.

Therefore, a lightweight supervised prior (created by training over the Kaggle dataset) should stabilize country/region guesses, with the LLM used only when the prior is uncertain.

2.1 Experimental Design

Upon inspection of the Kaggle dataset, we note that the dataset contains more information than is tested in evaluation. We undertake the following design choices in our implementation of an improved RAG system:

- **Data prep:** We download the Kaggle recipe CSV; we use the normalized ingredients and recipe names, character-level n-grams (3-5)
- **Models:** We run a logistic regression model to predict the country (`max_iter=300`, `multi_class='auto'`). The prior built only factors in country, not regions, so direct paths will be weak.
- **Inference logic:** We compute the probabilities of the top 3 possible countries. If the the topmost country has probability ≥ 0.60 , we return that country with region=All. Otherwise, we call a deterministic LLM (temperature=0) with a short prompt (containing the description, the top 3 country candidates with their probabilities, and the allowed regions)
- **Evaluation:** Score only on Culinary Detective entries in public.jsonl (val). If improved vs baseline, regenerate predictions for public.txt and then private.txt for final submission; private is held-out test.
- **Ensemble:** We will also test a combination of the improvements to create an ensemble, one part boosting the other.

We will implement two primary improvements:

- **Structured RAG:** Clean the Kaggle recipe corpus, build dense FAISS (and optional BM25 to improve RAG ranking), retrieve top snippets, and prompt a deterministic LLM with the snippets to emit [country], [region].
- **Logistic Regression prior:** Train a country classifier on labeled Kaggle data (the cuisine column of the dataset corresponds to the country) using char-level TF-IDF features and Logistic Regression. If the classifier is confident, emit its country, with region All. Otherwise, pass the top country candidates to a deterministic LLM to pick [country], [region] with region constrained to the allowed set.

We expect these improvements to lead to cleaner, more focused retrieval, which will reduce country drift. We will validate each stage of this implementation, building on itself, and report the results, submitting the best performing stage. Prior work shows RAG quality is gated by corpus relevance (Lewis et al., 2020; Cuconasu et al., 2024). Additionally, a TF-IDF + LR prior avoids noisy retrieval and grounds the LLM with candidate narrowing and explicit region constraints.

3 Experimental Results

To move beyond simply the scores, we added country-level confusion matrices to see where each system succeeds or fails by class. By plotting the gold vs. predicted countries for the top-15 (tail bucketed as “Other”), we can spot systematic confusions versus random noise, comparing how each approach concentrates mass on the diagonal. This allows us to understand whether gains in the score table come from fixing specific countries or just shifting errors elsewhere.

Each panel in the figure is a country confusion matrix over the top-15 gold countries (tail bucketed as “Other”). In each matrix, the rows correspond to the gold, and the columns correspond to the predicted labels. Additionally, a darker diagonal corresponds to higher recall for that gold country, and off-diagonal blocks mean systematic confusions.

For each of the iterative improvements we make on our baseline, we note the following results (and analyze them here):

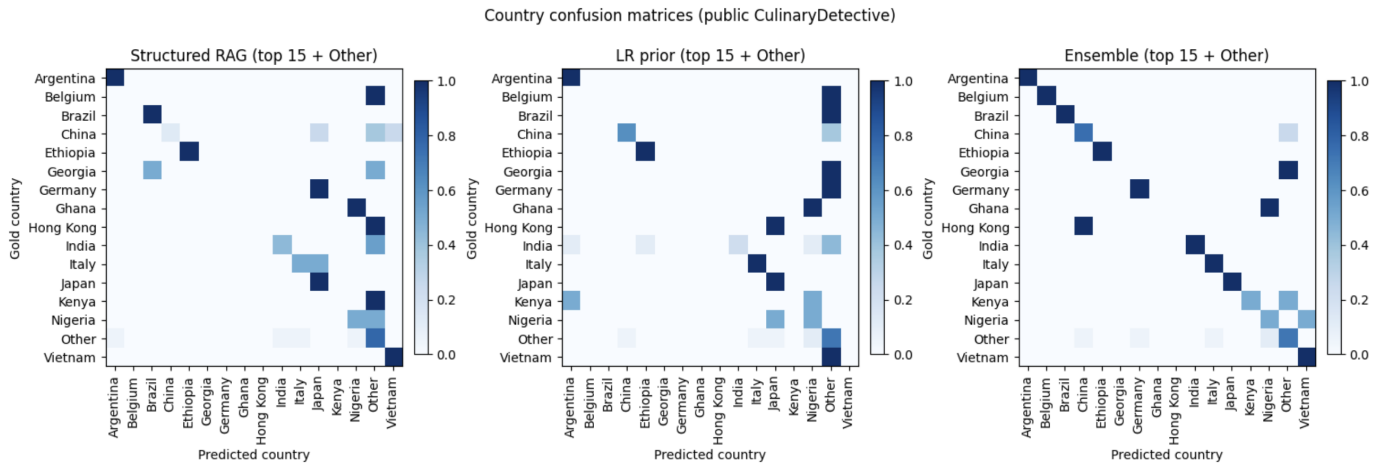


Figure 1: Country Confusion Matrices for Three Improvement Iterations

Here, we analyze these results:

- **Structured RAG:** diagonal is weak and spread as evidenced by the figure. The dispersion of the implementation matches the low score (0.21) and the noisy retrieval seen in our table. Gold Belgium often goes to Other, for instance. The retrieval noise pushed the LLM off the expected results, as the reranking implemented by using the BM-25 algorithm was not strong enough to anchor to the right country.
- **LR prior:** we see a similarly weak diagonal as in the structured RAG implementation, and we see further that many rows bleed into "Other". Accuracy here is still low (0.21), but it is much faster (29.6s compared to the 126.2s of the structured RAG implementation). The pseudo-labeling of the Kaggle training data set (mapping cuisine to country, and region to All), yields a rough prior that underfits many cuisines.
- **Ensemble:** this matrix shows the strongest diagonals overall, and most of the top-15 classes show high recall. We see some residual confusion still clusters around Nigera/Other, but far fewer off diagonals than the other two implementations. This aligns with the score results we have seen (0.43), though at the cost of latency (194.6s). The Ensemble’s gain comes from complementary errors: the zero-shot is generally strong, and LR occasionally nails ingredient-heavy classes, and the RAG helps in cases where snippets match.

Upon running our baseline and improvement iterations on our public set, we find our improvement ideas produce the results below.

Approach	Public set		Private set	
	Score	Time	Score	Time
Baseline	0.3973	45.62s		
Structured RAG	0.2109	126.2s		
Logistic Regression prior	0.2073	29.6s		
Ensemble	0.4327	194.6s	NA	207.1s

We note that the we could not accurately measure the score of the private set, and so we have only recorded the runtime of our strongest implementation, which we use to generate the public.txt and private.txt.

4 Analysis

We now dig into how our Culinary Detective systems behave beyond a single score. We focus on (1) on class-specific errors via country confusion matrices (Fig. 1, in the prior section) to see which cuisines are systematically confused; (2) the accuracy–efficiency frontier (Fig. 2) to show the cost/quality trade-offs among the baseline and our improvements; (3) regional bias via gold vs. predicted region frequencies (Fig. 3) to reveal trends of certain regions. Our analysis uses the public set as validation only, while the private set is held out. We aim to explain why some approaches win (ensemble) or lose (RAG/LR).

In our confusion matrices, we plotted the top-15 classes for the Structured RAG, Logistic Regression prior, and Ensemble iterations. From this, we note that across models, we see confusion in the Ghana gold/Nigera predicted, Georgia gold/Other predicted categories specifically. These likely arise from overlapping ingredients (tomato/onion/garlic) and generic descriptors. The ensemble reduces but doesn’t eliminate these clusters completely, suggesting a need for stronger discriminative signals (spices, prep styles) or a stronger region prior.

Additionally, we see both the RAG and LR lean heavily on the "Other" section, indicating low confidence on rare cuisines. The ensemble keeps "Other" usage lower by leveraging the zero-shot priors and LR hits, but rare classes still remain a weak spot. We hypothesize that the cause is a limitation of our pseudo-labeled training and noisy retrieval.

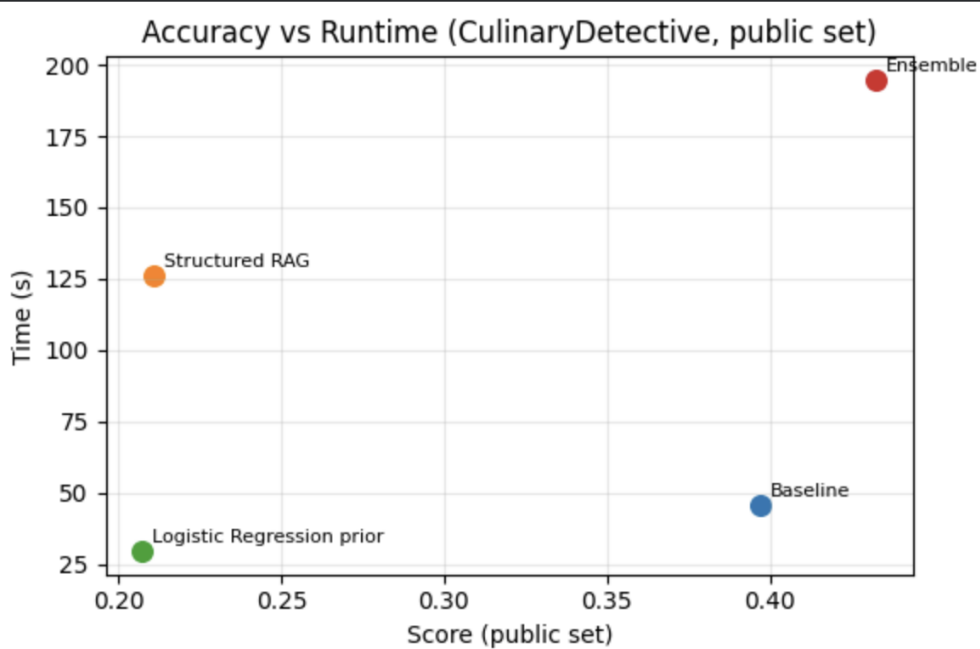


Figure 2: Accuracy-efficiency graph for the baseline and all improvement iterations

The accuracy–efficiency plot above makes the tradeoff explicit. The baseline sits with a 0.40 score in 46s. The LR prior is the fastest (30s) but yields the lowest accuracy (0.21), demonstrating that a lightweight pseudo-labeled prior alone underfits. Structured RAG is strictly dominated: slower (126s) and still 0.21 accuracy, illustrating that noisy retrieval hurts both quality and latency. The ensemble delivers the best accuracy (0.43) but at the highest cost (195s), roughly 4x the baseline, because it runs multiple heads. Practically, we choose between baseline (solid quality, moderate cost) and ensemble (highest quality, highest cost); our figure shows there is no free gain in this setup.

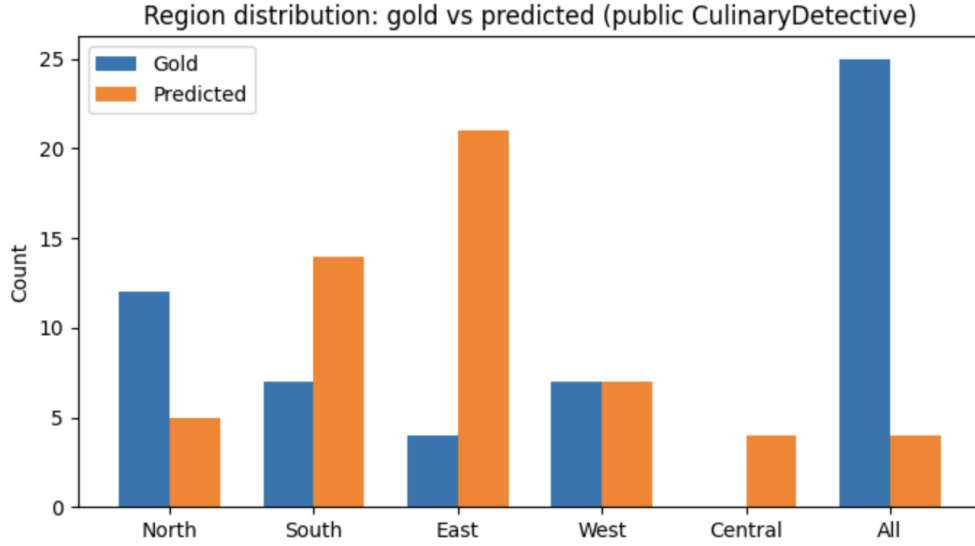


Figure 3: The region distribution gold vs prediction for the ensemble implementation

Figure 3 compares gold vs. predicted region counts of the ensemble situation. The model overuses East and South and under-predicts North and especially All (gold has a large All bucket the model rarely emits). Central appears spuriously despite no gold Central examples even with prompts explicitly demanding the model respond with only one of the restricted set, indicating region hallucination. West is roughly matched. Because region is exact-match scored, the skew toward East/South and the underuse of All directly depress region accuracy. This mismatch suggests our region signal is weak: pseudo-labeled training with region defaulted to All and the LLM still drift to directional regions. A better region prior, possibly by labeling a heuristic region per country during pseudo-training, or a more balanced training data, would likely reduce these systematic region errors.

Overall, the analysis shows that retrieval was hurt without a clean corpus, a weak prior alone underfits, and combining complementary heads yields the best accuracy at a latency cost. Confusion matrices highlight persistent clusters, while the region plot reveals overuse of East/South and underuse of All, driving exact-match losses. The accuracy–runtime scatter makes the trade-offs explicit: baseline is balanced, LR is cheap but weak, ensemble is best but slow. If we iterated further, we’d target better region priors and a leaner ensemble to approach the baseline’s runtime. All conclusions are drawn on the public (validation) split; private remains the held-out test for final reporting.

References

We utilized the following prompts for each of the sections. Multiple tweaks were made to try to improve the prompts, with only marginal improvements:

Global Trekker:

```
global_trekker_messages =
[
{"role": "system",
 "content": "You are a precise world geography expert. Return only the most
            likely country and city."},
{"role": "user",
 "content": "Given the paragraph, output exactly one line:
            [country], [city]. Do not add explanations or extra text.
            If no city is implied, write None for the city."},
]
```

Culinary Detective:

```
culinary_detective_messages =
[
{"role": "system",
 "content": "You are an expert in world cuisines and regional foods."},
{"role": "human",
 "content": "Given the description, return in one line the country and region
            (must be only ONE of {North, South, East, West, Central}). Answer
            only in one line, in brackets and separated by a comma: [country], [region].
            Do not add any other text."},
]
```

Lingua Locale:

```
lingua_locale_messages =
[
{"role": "system",
 "content": "You are an expert in official languages of government websites."},
{"role": "user",
 "content": "Given the sentence from an official website, determine which country's
            website it comes from. Answer in one line with no other text the country's
            name in brackets: [country]"},
]
```

Referenced Literature:

- Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (Lewis 2020)
- The Power of Noise: Redefining Retrieval for RAG Systems (Cuconasu, 2024)
- “How to Improve RAG Performance” (reddit.com/r/LangChain)